Renewable Energy 50 (2013) 188–195

Contents lists available at SciVerse ScienceDirect

Renewable Energy

journal homepage: www.elsevier.com/locate/renene

Comparison of physicochemical features of biooils and biochars produced from various woody biomasses by fast pyrolysis

Kwang Ho Kim [a,1], Tae-Seung Kim [a], Soo-Min Lee [b], Donha Choi [b], Hwanmyeong Yeo [a], In-Gyu Choi [a], Joon Weon Choi [a,*]

[a] Department of Forest Sciences and Research Institute for Agriculture and Life Science, Seoul National University, Gwanak-ro, Gwanak-gu, Seoul 151-921, Republic of Korea
[b] Div. of Wood Chemistry & Microbiology, Korea Forest Research Institute, 57 Hoegi-ro, Dongdaemun-gu, Seoul 130-712, Republic of Korea

ARTICLE INFO

Article history:
Received 18 May 2011
Accepted 18 June 2012
Available online 20 July 2012

Keywords:
Lignocellulosic biomass
Fast pyrolysis
Biooil
Biochar
Fluidized bed reactor

ABSTRACT

The objective of this study was to characterize biooils and biochars produced from four different woody biomasses (two hardwoods: oak, eucalyptus and two softwoods: pitch pine, Japanese cedar). Fast pyrolysis was performed at 500 °C for ca. 2 s using fluidized bed reactor. The mass balances of essential pyrolytic products (biooil, biochar and gas) were influenced by the types of feedstocks. The yields of biooils and biochars were determined to ca. 62–68% and 11–14% based on dry weight of feedstock, respectively. However, physical properties of biooils such as pH, water content, and heating value were almost similar regardless of feedstocks. All biooils were acidic (pH 1.7–2.4) and water contents were estimated between 20% and 26%. The heating values were determined to 15.5–19 MJ/kg. X-ray fluorescence (WDXRF) spectrometry and scanning electron microscopy analyses of biochars showed that considerable amounts of mineral components, which could play an important role for plants growth, were detected in the biochars.

1. Introduction

Biomass, as the only source of renewable carbon, shows great promise for the large-scale economical production of renewable energy and chemicals. In addition, biomass is an extremely abundant resource that can be produced in agriculture, forestry and microbial systems [1]. However, a proper energy conversion process is necessary to utilize the biomass as a renewable energy resource because of its recalcitrance. This trait which comes from compositional resistance of biomass to deconstruction from microbes and enzymes has made it difficult to be used as an energy resource.

There are two main processes to convert biomass into fuels and chemicals; biochemical and thermochemical processes [2]. Thermochemical processes are classified into pyrolysis, gasification and combustion. Among the three main thermochemical processes, pyrolysis is one of the promising technologies for liquefaction of biomass [3–5]. Fast pyrolysis, characterized *by simple process* and short reaction time, has a capability of transforming the bulky and inhomogeneous biomass to liquid form of fuel (which is called to biooil). Besides, wide applicability of liquid biooil not only as an alternative energy but also green chemical resources is highly regarded. In addition to biooil, the biochar, a co-product of pyrolysis, has attracted great attention. The biochar is potentially a valuable soil amendment because it contains most of the mineral nutrients, which is a good absorbent of nutrients and agricultural chemicals and may sequester carbon [6]. Utilization of biochar as a soil amendment will offset many problems associated with removing crop residues from the land [6]. Soil sequestration of carbon through biochar offers a means of mitigating climate change [7,8]. Also, the biochar may be used *as a carbon-rich* solid biofuels [9,10]. A lot of studies regarding biooil and biochar production from various biomass sources such as woody biomasses [10–14], agricultural residues [9,15–18], energy crops [19,20] and animal wastes [21–24] have been reported.

For the comparative study for physicochemical properties of biooils and biochars, especially compositional analysis of biooil as well as morphological feature of biochars, in terms of wood species, oak, eucalyptus, pitch pine and Japanese cedar were chosen as raw materials of fast pyrolysis and subjected to fast pyrolysis using fluidized bed reactor at 500 °C with residence time of pyrolysis products of ca. 2s, which experimental condition was established as an optimal condition for biooil production by previous study [25].

* Corresponding author. Tel.: +82 2 880 4788; fax: +82 2 873 2318.
E-mail address: cjw@snu.ac.kr (J.W. Choi).
[1] Present address: Center for Sustainable Environmental Technologies, Iowa State University, United States.

0960-1481/$ – see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.renene.2012.06.030

Table 1
Compositional analyses of raw materials.

Properties (wt%, dry basis)	Oak	Eucalyptus	Pitch pine	Japanese cedar
Elemental analysis				
Carbon	45.7	46.4	46.8	48.8
Hydrogen	6.0	5.9	6.0	6.1
Nitrogen	<0.2	<0.2	<0.2	<0.2
Oxygen[a]	48.3	47.6	47.1	45.1
Components analysis				
Holocellulose	80.0	81.4	76.1	73.3
Lignin	24.7	25.6	28.4	35.1
Extractives	2.8	1.5	6.9	3.6
Ash	0.8	0.4	0.5	0.4
Calorific value (MJ/kg)	17.8	16.5	17.9	19.2

[a] By difference.

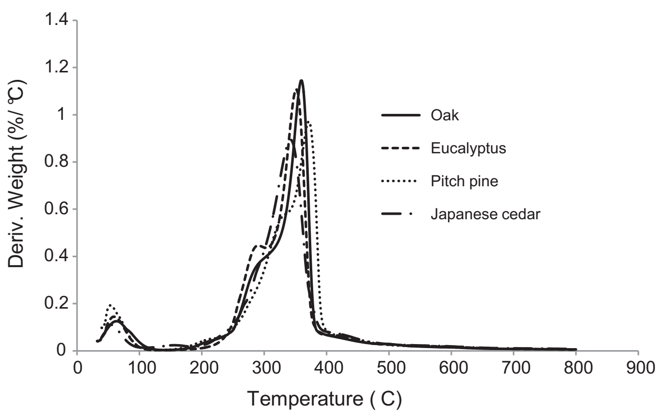

Fig. 1. Thermogravimetric analysis (DTG curves) of various lignocellulosic biomasses.

2. Experimental

2.1. Feedstocks

As raw materials of fast pyrolysis, xylem tissues of oak, eucalyptus, pitch pine and Japanese cedar were prepared. Samples were air dried to about 5% moisture content prior to pyrolysis. Debarked samples were ground and screened to pass a screen of 0.5 mm aperture. *Elemental analysis (CHNS-932 of LECO Corp.) and component analysis were carried out by NREL procedures* [26] *and summarized in* Table 1. Before pyrolysis in the fluidized bed type fast pyrolysis system, thermogravimetric analysis (TGA) was applied by Q-5000IR of TA Instruments coupled with differential thermal analyzer (DTA) to evaluate the pyrolytic behavior of raw materials. For TGA experiment 30 mg of sample was used with 10 ml/min heating rate. Nitrogen was used as a carrier gas with flow rate of 25 ml/min.

As shown in Fig. 1, the weight loss up to 150 °C for all samples was linked to the release of water from raw materials. And main components, cellulose, hemicelluloses and lignin, were degraded from 250 °C to 600 °C. Thermal decomposition of lignocellulosic biomasses, *as reported earlier* [27,28], begun about 250 °C with decomposition of hemicelluloses and successive decompositions of cellulose and lignin took place at 300–600 °C. The maximum thermal decomposition rate of all raw materials *was observed at about 350 °C.*

2.2. Performance of fast pyrolysis

Fast pyrolysis of various feedstocks was performed at 500 °C and 2 s of residence time using fluidized bed reactor (400 mm × 35 mm), including feeder, cyclone for char separation followed by two condensers maintained at about 0 °C and an electrostatic precipitator for collection of biooil. *A schematic diagram of the fluidized bed reactor is shown in* Fig. 2. Before experiment the reactor was purged for 30 min by nitrogen to remove air. *150g of each sample was fed into reactor for 1 h under pyrolytic condition*. With the completion of fast pyrolysis, the yields of biooil and biochar were determined from overall material mass balance analysis. Then, the produced biooil was refrigerated immediately at 4 °C to prevent aging.

2.3. Analytical methods

Several physical and chemical analyses were conducted to *characterize the obtained biooil* and biochar. The list of analytical methods for biooil is shown in Table 2. Water content, acidity, viscosity, calorific value and elemental composition of biooils were analyzed respectively by analytical methods. In addition, GC/MS analysis was carried out to identify individual chemical compounds in biooils. The quantitative analysis for biooil components was conducted by comparing peak area of internal standard detected by FID. Fluoranthen (Sigma) was used as an internal standard (50 mg/

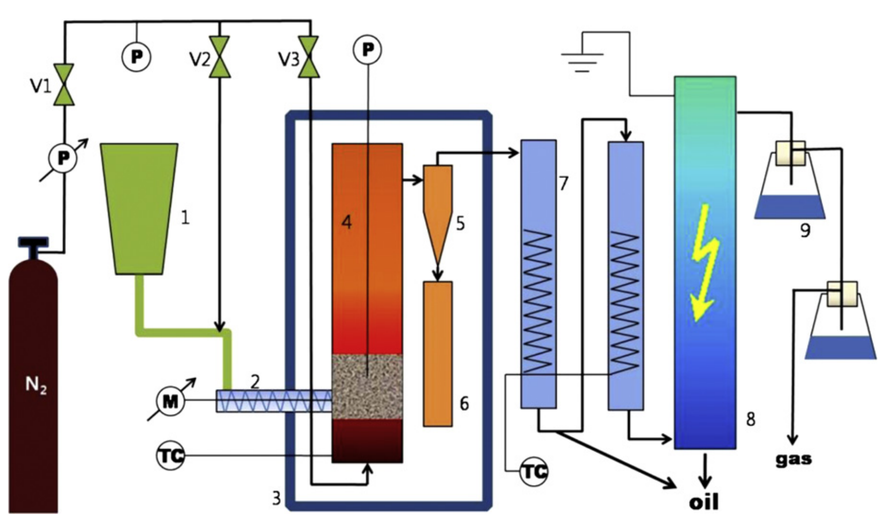

Fig. 2. Schematic diagram of fluidized bed reactor (1: feeder, 2: screw feeder, 3: isolation, 4: reactor, 5: cyclone, 6: char collector, 7: cooler, 8: electrostatic precipitator, 9: filter).

Table 2
Analytical methods of biooil.

Property	Method	Unit
Water content	Karl-Fischer titration	%(w/w, wet basis)
pH	pH meter	pH unit
Viscosity (@ 40 °C)	Capillary type viscometer	cSt
Calorific value	Bomb type calorimeter (Parr 6400, Parr instrument company)	MJ/kg
Elemental analysis	Elemental analyzer (CHNS-932 Leco Corp.)	%C, %H, %N (w/w, wet basis)
Gas chromatography	Column type: DB 5 Dimensions: 30 m × 0.25 mm Film thickness: 0.25 μm Injector: 250 °C, split ratio 1:20 FID detector: 280 °C Oven: 45 °C (4 min. hold), 3 °C/min to 280 °C (20 min. hold)	

2 mL acetone). 50 μL of fluoranthen solution *was added to each* biooil sample.

The elemental analysis of the biochars was performed by using CHNS-932 elemental analyzer of Leco Corp. The surface morphology of the biochars was investigated by a field-emission scanning electron microscope (FE-SEM, SUPRA 55VP of Carl Zeiss). For analysis of FE-SEM, the samples were mounted on aluminum stubs using carbon tape with conductive silver paint applied to the sides to reduce sample charging and sputter-coated with Pt-Pd by sputter coater. Imaging was performed at 3 kV of beam voltages. X-ray fluorescence (WDXRF) spectrometry (S4 PIONEER, Bruker AXS Co.) was applied to identify and quantify inorganic metals in biochars. For this analysis, 3 g of sample of the biochars was taken and the mixed with binder (a high quality paraffin material), and pressed to make a tablet like specimen. During XRF testing, *X-rays* were used to excite the electrons of the elements in the specimen. Then the excited electron underwent energy decay, and the energy released was captured and analyzed by the XRF apparatus. The frequency, time delay, and other factors of this fluorescent energy led to the determination of the elemental make up of the samples.

3. Results and discussion

3.1. Products distributions

It was reported in the previous study [25] that the maximum yield of biooil (ca. 68%, based on dried biomass weight) was obtained from yellow poplar sawdust at 500 °C of pyrolysis temperature and 1.9 s of residence time of pyrolysis products. Under the same experimental condition the yields of biooils were determined to 65.7%, 59.2%, 61.6% and 62.6% of the feedstock for oak, eucalyptus, pitch pine and Japanese cedar respectively. Biochar yields were ranged between 13.9% and 16.5%, while gases were measured to 20.2%–25.9% depending on the feedstocks. As can be seen in Table 3, there is no clear difference in biooil yield between softwood and hardwood. The yields of biochar and gas, similar to those of biooil, were not different between wood species, either. From the results of pyrolysis products distribution of four species, fast pyrolysis process has no material specificity for biooil and biochar production. Interestingly, the amount of water newly formed during pyrolysis in biooils from oak and eucalyptus was higher than that in biooils from pitch pine and Japanese cedar. It means that dehydration reaction could occur more dominantly in hardwood compared to softwood during pyrolysis. *This is because hemicellulose in hardwoods contains more highly acetylated form than in softwoods* [29]. When pyrolyzed, hemicellulose is the most susceptible to thermal decomposition among the three main components of biomass and easily liberates water from it changing into furfural.

Table 3
Products (biooil, biochar, gas and water) distribution.

wt. % (dry basis)	Oak	Eucalyptus	Pitch pine	Japanese cedar
Biooil	65.7	59.2	61.6	62.6
Biochar	14.1	14.9	16.5	13.9
Gas	20.2	25.9	21.9	23.5
Water formation	9.2	9.6	6.5	7.6

3.2. Properties of biooils

The characteristics of pH, water content, viscosity, calorific values and elemental analysis for the biooils are summarized in Table 4. As shown in this Table, the water content of all biooils ranged from 15.2 to 26.4%. This means that significant amount of water was produced during thermal degradation of biomass. Water produced resulted from original moisture in feedstock and dehydration reactions during fast pyrolysis [30]. *From the point of view of fuel, the presence of water in biooil can reduce viscosity and ensure a uniform temperature distribution in chamber* [31]. *However, the presence of water in biooil results in several problems such as low heating value* and phase separation of biooil. Since various acids such as acetic acid and vanillic acid were produced during pyrolysis, *pH value of biooil was between 1.7 and 2.4*. Biooils from softwood were high in carbon, which led higher calorific value of 18.4 MJ/kg and 18.9 MJ/kg for pitch pine and Japanese cedar, respectively in comparison with biooils from hardwood (17.0 MJ/kg for oak tree and 15.5 MJ/kg for eucalyptus). When comparing the heating value of conventional fossil fuel, the low heating values of biooil were attributed to the *high water and oxygen content. For this reason, the quality of biooil should be improved by catalytic upgrading to be suitable for transportation fuel.*

3.3. Compositions of biooils

Since lignocellulosic biomass is composed of cellulose, hemicellulose and lignin, the major pyrolysis products of woody material are derived from these three components with different proportions [32]. It has been reported that more than 300 chemical components exist in biooil [30]. But complete identification of all chemical components is impossible by chromatographic analysis due to biooil complexity. In this research, chemical composition of biooils was analyzed by gas chromatographic method equipped with FID and MS detector. More than 70 peaks appeared on the chromatogram. Among them, ca. 20 compounds were identified by comparison of mass spectra of each peak with those of authentic compounds in the National Institute of Standards and Technology (NIST) mass spectra library, then semiquantified with FID by

Table 4
Characteristics of biooils.

	Oak	Eucalyptus	Pitch pine	Japanese cedar
Water content (wt%)	20.2	26.4	23.6	20.5
pH	2.4	1.7	2.5	2.4
Viscosity (cSt, @ 40 °C)	34.6	19.4	10.3	45.9
Elemental analysis (wt%)				
Carbon	41.4	38.5	44.1	50.1
Hydrogen	7.1	6.8	6.6	6.7
Oxygen[a]	51.3	54.6	49.1	43.0
Nitrogen	n.d.			
HHV (MJ/kg)	17.0	15.5	18.3	18.9

[a] By difference.

addition of fluoranthen as an internal standard. Table 5 and Fig. 4 show the list of compounds identified and their essential chemical structures, respectively. As can be seen in this Table, each biooil was composed of a number of compounds derived from carbohydrates and lignin.

2-Furaldehyde (2), 2-butanone (3), 2(5H)-furanone (6), 3,5-dihydroxy-2-methyl-4H-pyran-4-one (10) and 1,6-anhydro-β-D-glucopyranose (levoglucosan; 11) were some degradation products of carbohydrate pyrolysis. Levoglucosan was the major chemical compound in biooils derived from thermal degradation of cellulose. It was reported that levoglucosan was produced from the intramolecular transglycosylation of cellulose during pyrolysis [33]. Among four feedstocks the highest amount of levoglucosan was detected in the biooil produced from eucalyptus, while trace amount of levoglucosan was detected in the biooil from Japanese cedar. As shown in Fig. 3, since the signal for levoglucosan on the chromatogram appeared broadly at retention time between 42min and 46min and overlapped unfortunately with other signals, the exact quantification of levoglucosan amount was still difficult.

Lignin pyrolysis, although it depends on raw materials, generates numerous phenolic compounds. In hardwood biooil, 4-hydroxy-3-methoxybenzoic acid (vanillic acid; 19), 2,6-dimethoxy-4-(2-propenyl) phenol (methoxy eugenol; 26) and 2,6-dimethoxy phenol (syringol; 23) were main coumpounds derived from lignin. In case of softwood biooil, 3-methoxy-4-methylphenol (3-methoxy-*p*-cresol; 15), 4-hydroxy-3-methoxybenzoic acid (vanillic acid; 19) and 3-methoxy-4-vinylphenol (4-vinylguaiacol; 16) were highly concentrated in softwood biooil. Actually, syringyl group cannot be involved in lignin synthesis of softwood, components and contents of pyrolysis products from lignin were clearly different between softwood and hardwood. Structures of essential pyrolytic products derived from carbohydrate and lignin are present in Fig. 3.

The *values* in the Table 5 were expressed with the area ratios between components in the biooil and a certain amount of internal standard added in the biooil before GC analysis (fluoranthene 1.25 mg). The sum of each fraction was multiplied by individual yield of biooil and the values in parentheses in Table 5 indicated the plausible portion of each fraction in the biooil. Concentration of carbohydrates and lignin derivatives in oak and eucalyptus was higher than that in pitch pine and Japanese cedar. As mentioned above, the concentration of levoglucosan in eucalyptus took approximately 75% of GC detectable compounds.

Compared to the chromatograms of biooils in Fig. 3, it can be easily observed that lots of signals appeared newly at retention time after 40 min in the chromatogram of biooil produced from Japanese cedar. According to mass spectral data analysis, these signals were identified as typical natural compounds found in Japanese cedar [34], such as δ-cadinene, δ-selinene, eudesmol and ferruginol. The structures of these compounds are present in Fig. 5, and the concentrations of each compound are given in Table 6. During severe pyrolysis condition, some of these compounds without thermal decomposition were transferred to biooil phase. Since these extractives are well known to strong antifungal activity [34,35], biooil from Japanese cedar can be utilized not only as fuel but also as antifungal agents such as wood preservatives.

Table 5
Identification and semiquantitative analysis of low molecular weight compounds in the biooils produced from different wood species (H: *p*-hydroxyphenyl types, G: guaiacyl types and S: syringyl types).

		Compounds	Concentration (area/IS area)			
			Oak	Eucalyptus	Pitch pine	J. cedar
Carbohydrates		1. Acetic acid	0.73	0.99	0.36	0.52
		2. 2-Furaldehyde	0.19	0.12	0.01	0.01
		3. 2-Butanone	0.83	0.36	0.51	0.45
		4. 1-Acetyloxy-2-propanone	0.54	0.18	0.28	0.68
		5. 4-Cyclopentene-1,3-dione	0.20	0.07	0.22	0.25
		6. 2(5H)-Furanone	0.76	1.90	0.49	0.62
		7. 3,4-Dihydro-2H-pyran-2-one	0.21	0.21	0.14	0.16
		8. 1,2-Cyclopentanedione	0.12	0.09	0.07	0.10
		9. 2-Hydroxy-3-methyl-2-cyclopenten-1-one	0.67	1.52	0.38	0.23
		10. 3,5-Dihydroxy-2-methyl-4H-pyran-4-one	0.37	0.41	0.28	0.72
		11. 1,6-Anhydro-ß-D-glucopyranose (Levoglucosan)	1.31	8.24	1.78	0.33
Sum of carbohydrates			5.9(3.9)[a]	14.1(8.3)[a]	4.5(2.8)[a]	4.1(2.6)[a]
Lignin	H unit	12. Phenol	0.05	0.06	0.05	0.08
		13. 4-Methyl-phenol (*p*-cresol)	0.09	0.05	0.05	0.11
		Sum of H unit	0.14	0.11	0.10	0.19
	G unit	14. 4-Methoxyphenol (*p*-Guaiacol)	0.05	0.08	0.07	–
		15. 3-Methoxy-4-methylphenol (2-methoxy-*p*-Cresol)	0.52	0.41	0.94	0.95
		16. 3-Methoxy-4-vinylphenol (4-vinylguaiacol)	0.22	0.24	0.48	0.65
		17. 3-Methoxy-4-(2-propenyl)-phenol (eugenol)	0.08	0.17	0.32	0.48
		18. 4-Hydroxy-3-methoxybenzaldehyde (vanillin)	0.07	0.23	0.38	0.61
		19. 4-Hydroxy-3-methoxybenzoic acid (vanillic acid)	0.99	0.49	0.73	1.12
		20. 3-Methoxy-4-(1-propenyl)-phenol (isoeugenol)	0.13	0.50	0.33	0.57
		21. 3-Methoxy-4-propyl phenol (4-propyl guaiacol)	0.08	0.15	0.10	0.13
		22. 4-Hydroxy-3-methoxycinnamaldehyde (coniferaldehyde)	0.04	–	–	–
		Sum of G unit	2.18	2.28	3.36	4.50
	S unit	23. 2,5-Dimethoxy phenol (syringol)	0.92	0.70		
		24. 2,3,5-Trimethoxytoluene	0.16	0.11		
		25. 4-Hydroxy-2,5-dimethoxy benzaldehyde (syringealdehyde)	0.15	0.24		
		26. 2,5-Dimethoxy-4-(2-propenyl) phenol (methoxy eugenol)	0.89	0.95		
		27. 1-(4-hydroxy-3,5-dimethoxyphenyl) ethanone (acetosyringon)	0.19	0.19		
		28. 3,5-Dimethoxy-4-hydroxyphenylacetic acid (homosyringic acid)	0.21	0.14		
		29. 3,5-Dimethoxy-4-hydroxycinnamaldehyde (synapylaldehyde)	0.07	0.06		
		Sum of S unit	2.58	2.39		
	Sum of Linin		4.9(3.2)[a]	4.8(2.8)[a]	3.5(2.2)[a]	4.7(2.9)[a]
Total			10.8(7.1)[a]	18.9(11.1)[a]	8.0(5.0)[a]	8.8(5.5)[a]

[a] Multiplying the mass fraction of each compound by biooil yield (See Table 3), indicating the plausible portion in the biooil.

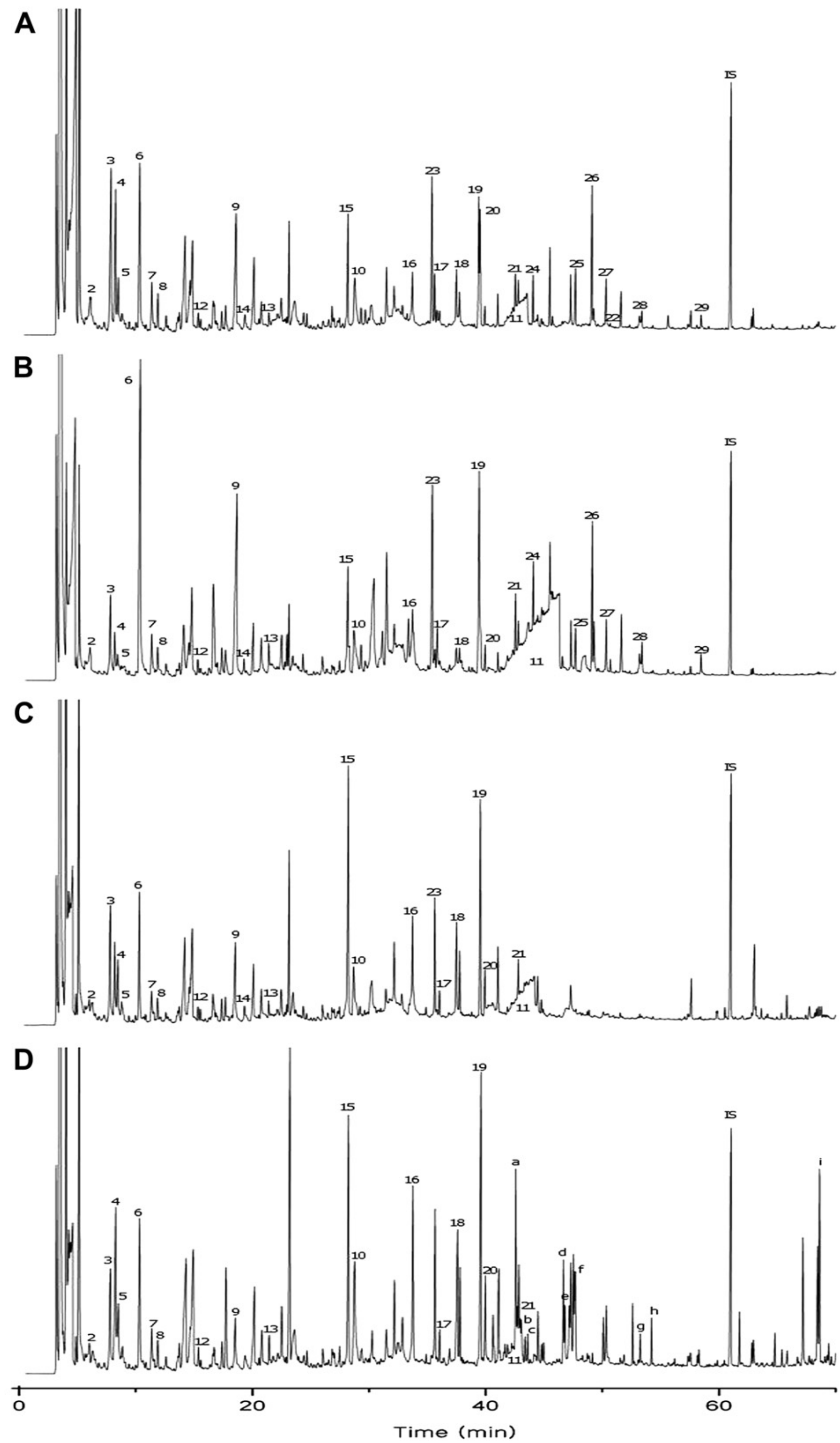

Fig. 3. Gas chromatographic separation of pyrolytic products in the various biooils (A: oak, B: eucalyptus, C: pitch pine, D: Japanese cedar).

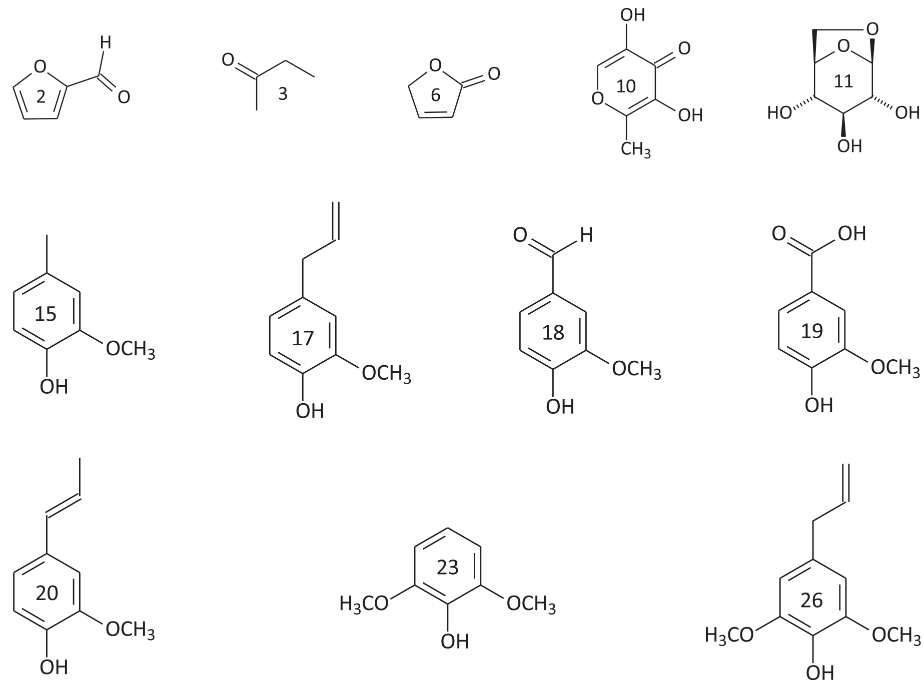

Fig. 4. Essential pyrolytic products derived from carbohydrate and lignin.

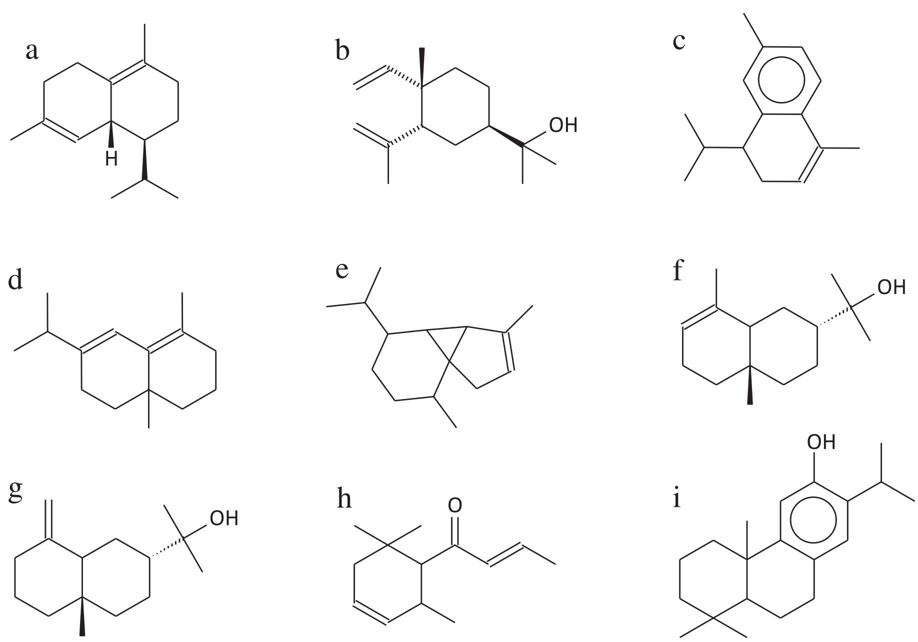

Fig. 5. Structures of natural compounds in the biooil produced from Japanese cedar.

Table 6
Natural compounds detected in the biooil produced from Japanese cedar.

Compounds (wet basis)		Concentration (area/IS area)
a.	δ-Cadinene	0.52
b.	Elemol	0.20
c.	Calacorene	0.13
d.	δ-Selinene	0.29
e.	α-Cubebene	0.19
f.	α-Eudesmol	0.43
g.	β-Eudesmol	0.08
h.	δ-Damascone	0.13
i.	Ferruginol	0.60
Total		2.57 (1.5)[a]

[a] Multiplying the mass fraction of each compound by biooil yield (biooil yield: 59.2%).

Table 7
Elemental composition and calorific value of biochars.

	Oak	Eucalyptus	Pitch pine	Japanese cedar
Elemental analysis (wt%, dry basis)				
Carbon	85.9	90.0	88.7	87.8
Hydrogen	1.4	1.4	1.4	1.4
Nitrogen	Trace			
Calorific value (MJ/kg)	30.2	32.2	31.5	31.2

Table 8
Inorganic metals in biochars (XRF analysis).

g/kg	Oak	Eucalyptus	Pitch pine	Japanese cedar
Si	33.34	4.22	22.87	30.63
K	16.57	2.46	8.33	16.10
Ca	16.00	2.24	5.22	4.06
Al	8.63	0.71	6.12	7.16
Mg	4.83	0.87	2.10	1.29
Na	1.81	0.71	1.30	2.31
P	1.13	0.37	0.67	0.18
Mn	0.45	0.10	0.60	–
Fe	0.42	0.35	0.76	0.38

3.4. Biochars

Biochar produced as a co-product of biomass pyrolysis has attracted a lot of interest in terms of its wide applications in environment and industry. Although its yield is dependent on experimental condition, generally the yield of biochar produced in the fast pyrolysis process is around 13%. This biochar can be used as a solid fuel and soil amendment. Table 7 shows that elemental composition and calorific value of biochars. As shown in this Table, biochars have high calorific value approximately 31 MJ/kg, which is comparable value to coal. As a solid fuel, biochar has been tested in the form of pellet and briquette with agglomerating materials such as clay and starch [36].

Carbon content in wood is about 50%, whereas carbon in biochar accounts for 80–90% of it, which can be permanently sequestered in soil. Carbon in biochar is highly stable in soil environments and may be sequestered for thousands of years [6]. Also, biochar is able to lift soil and crop productivity. From the XRF data (Table 8), a number of inorganic metals were in biochar. Among them, potassium, phosphorus, calcium and magnesium were highly concentrated in biochar, which are important plant mineral nutrients. Morphological analysis of biochar was carried out by applying FE-SEM. FE-SEM images of both the feedstock and biochar are shown in Fig. 6. As can be seen in this Figure, the large structural differences have been occurred during fast pyrolysis. Biochar had been twisted out of shape. The SEM images also illustrated porous structure of biochar.

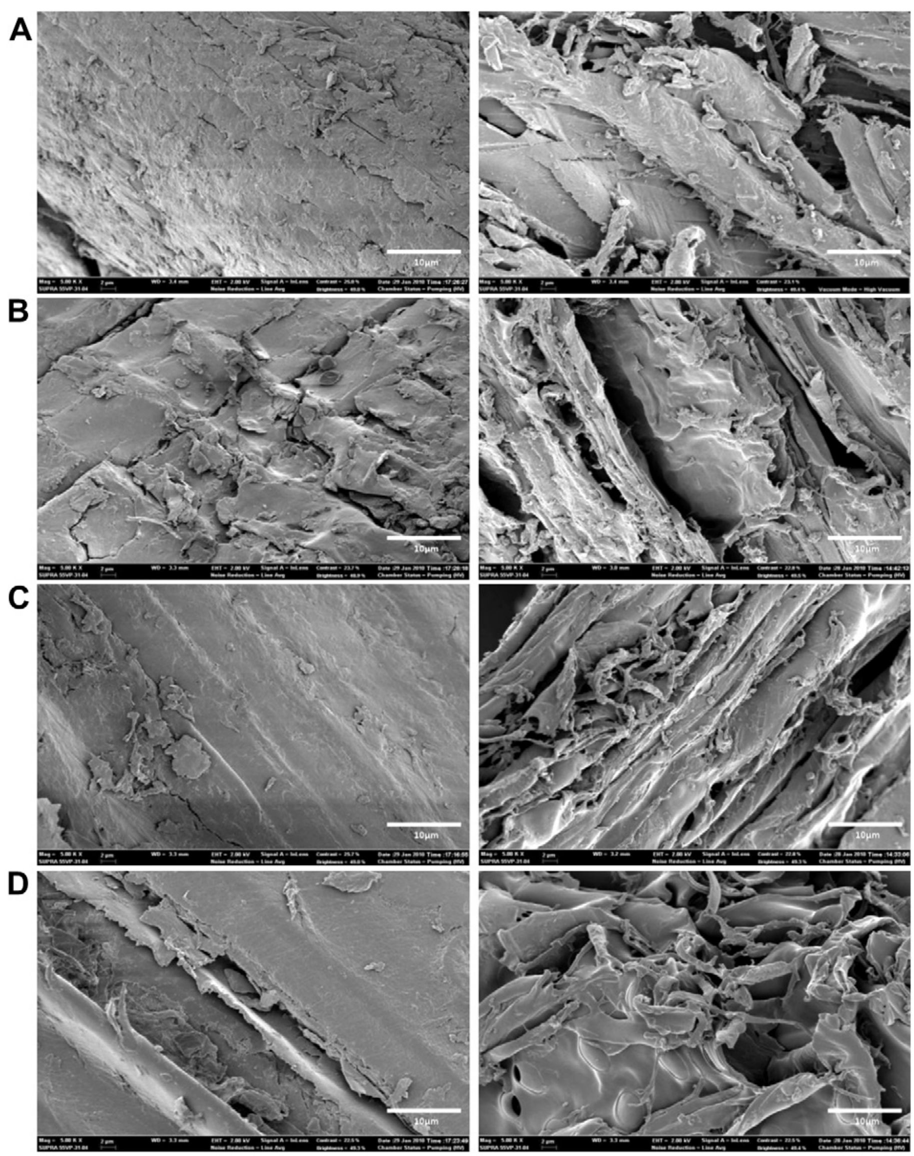

Fig. 6. FE-SEM images (×5000) of feedstocks (left) and biochars (right) (A: oak, B: eucalyptus, C: pitch pine, D: Japanese cedar).

4. Conclusions

In this study, fast pyrolysis of four different wood species (Oak, Eucalyptus, Pitch pine and Japanese cedar) was carried out to produce biooil and biochar in the fluidized bed reactor at constant pyrolysis temperature and residence time of volatiles (500 °C and 1.9 s). The yield of biooil ranged 59–65%. Although biooil yield was dependent upon raw materials, there was not clear difference in biooil yield between wood species. Calorific value of biooil of softwood was higher than that of hardwood, which was attributed to high carbon content. Difference in chemical compositions between raw materials was determined by gas chromatography system. The yield of biochars was 14–16% on mass basis. Biochars were characterized by elemental analysis, calorific value, XRF and SEM techniques. The biochar has been shown its potential for a natural fertilizer because it contains most of the important nutrition such as K, Ca, P and Mg for the plants grow. Lastly, biochar has shown the possibility of using as a solid fuel.

Acknowledgment

This study was carried out with the support of 'Forest Science & Technology Projects (Project No. S120811L140140)' provided by Korea Forest Service.

References

[1] Foust TD, Ibesen KN, Dayton DC, Hess JR, Kenney KE. The biorefinery. In: . Himmel ME, editor. Biomass recalcitrance. Oxford: Blackwell Publishing Ltd; 2008. p. 7–37.
[2] Karagoz S. Energy production from the pyrolysis of waste biomasses. Int J Energy Res 2009;33(6):576–81.
[3] Bridgwater T. Biomass for energy. J Sci Food Agric 2006;86(12):1755–68.
[4] Czernik S, Bridgwater AV. Overview of applications of biomass fast pyrolysis oil. Energy Fuels 2004;18(2):590–8.
[5] Bridgwater AV, Meier D, Radlein D. An overview of fast pyrolysis of biomass. Org Geochem 1999;30(12):1479–93.
[6] Mullen CA, Boateng AA, Goldberg NM, Lima IM, Laird DA, Hicks KB. Bio-oil and bio-char production from corn cobs and stover by fast pyrolysis. Biomass Bioenergy 2010;34(1):67–74.
[7] Winsley P. Biochar and bioenergy production for climate change mitigation. New Zealand Sci Rev 2007;64(1):5–10.
[8] Gaunt J, Lehmann J. Energy balance and emissions associated with biochar sequestration and pyrolysis bioenergy production. Environ Sci Technol 2008; 42(11):4152–8.
[9] Ozcimen D, Karaosmanoglu F. Production and characterization of bio-oil and biochar from rapeseed cake. Renewable Energy 2004;29(5):779–87.
[10] Demirbas A. Production and characterization of bio-chars from biomass via pyrolysis. Energy Sources Part A-Recov Utiliz Environ Effects 2006;28(5): 413–22.
[11] Balat M, Demirbas MF. Bio-oil from pyrolysis of Black Alder wood. Energy Sources Part A-Recov Utiliz Environ Effects 2009;31(19):1719–27.
[12] Boucher ME, Chaala A, Roy C. Bio-oils obtained by vacuum pyrolysis of softwood bark as a liquid fuel for gas turbines. Part I: properties of bio-oil and its blends with methanol and a pyrolytic aqueous phase. Biomass Bioenergy 2000;19(5):337–50.
[13] Schroder E. Experiments on the pyrolysis of large beechwood particles in fixed beds. J Anal Appl Pyrolysis 2004;71(2):669–94.
[14] Matsuoka K, Shinbori T, Kuramoto K, Nanba T, Morita A, Hatano H, et al. Mechanism of woody biomass pyrolysis and gasification in a fluidized bed of porous alumina particles. Energy Fuels 2006;20(3):1315–20.
[15] Gercel HF. Production and characterization of pyrolysis liquids from sunflower-pressed bagasse. Bioresour Technol 2002;85(2):113–7.
[16] Sanchez ME, Lindao E, Margaleff D, Martinez O, Moran A. Pyrolysis of agricultural residues from rape and sunflowers: production and characterization of bio-fuels and biochar soil management. J Anal Appl Pyrolysis 2009; 85(1–2):142–4.
[17] Sensoz S, Angin D. Pyrolysis of safflower (*Charthamus tinctorius* L.) seed press cake: part 1. The effects of pyrolysis parameters on the product yields. Bioresour Technol 2008;99(13):5492–7.
[18] Putun AE, Apaydin E, Putun E. Bio-oil production from pyrolysis and steam pyrolysis of soybean-cake: product yields and composition. Energy 2002; 27(7):703–13.
[19] He RH, Ye P, English BC, Satrio JA. Influence of pyrolysis condition on switchgrass bio-oil yield and physicochemical properties. Bioresour Technol 2009;100(21):5305–11.
[20] Yorgun S, Simsek YE. Fixed-bed pyrolysis of Miscanthus x giganteus: product yields and bio-oil characterization. Energy Sources 2003;25(8):779–90.
[21] Schnitzer MI, Monreal CM, Facey GA, Fransham PB. The conversion of chicken manure to biooil by fast pyrolysis I. Analyses of chicken manure, biooils and char by ^{13}C and ^{1}H NMR and FTIR spectrophotometry. J Environ Sci Health B 2007a;42(1):71–7.
[22] Schnitzer MI, Monreal CM, Jandl G, Leinweber P, Fransham PB. The conversion of chicken manure to biooil by fast pyrolysis II. Analysis of chicken manure, biooils, and char by curie-point pyrolysis-gas chromatography/mass spectrometry (Cp Py-GC/MS). J Environ Sci Health B 2007b;42(1):79–95.
[23] Dias BO, Silva CA, Higashikawa FS, Roig A, Sanchez-Monedero MA. Use of biochar as bulking agent for the composting of poultry manure: effect on organic matter degradation and humification. Bioresour Technol 2010;101(4): 1239–46.
[24] Lima IM, Boateng AA, Klasson KT. Pyrolysis of Broiler manure: char and product gas characterization. Ind Eng Chem Res 2009;48(3):1292–7.
[25] Kim KH, Eom IY, Lee SM, Choi D, Yeo H, Choi I-G, et al. Investigation of physicochemical properties of biooils produced from yellow poplar wood (Liriodendron tulipifera) at various temperatures and residence times. J Anal Appl Pyrolysis 2011;92(1):2–9.
[26] Sluiter A, Hames B, Ruiz R, Scarlata C, Sluiter J, Templeton D. Technical report. TP-510–42622. Golden, Co: National Renewable Research Laboratory; 2008.
[27] Di Blasi C. Modeling chemical and physical processes of wood and biomass pyrolysis. Prog Energy Combust Sci 2008;34(1):47–90.
[28] Mohan D, Pittman CU, Steele PH. Pyrolysis of wood/biomass for bio-oil: a critical review. Energy Fuels 2006;20(3):848–89.
[29] Palmqvist E, Hahn-Hagerdal B. Fermentation of lignocellulosic hydrolysates. II: inhibitors and mechanisms of inhibition. Bioresour Technol 2000;74(1): 25–33.
[30] Lu Q, Li WZ, Zhu XF. Overview of fuel properties of biomass fast pyrolysis oils. Energy Conversion Manage 2009;50(5):1376–83.
[31] Westerhof RJM, Kuipers NJM, Kersten SRA, van Swaaij WPM. Controlling the water content of biomass fast pyrolysis oil. Ind Eng Chem Res 2007;46(26): 9238–47.
[32] Hassan EBM, Steele PH, Ingram L. Characterization of fast pyrolysis bio-oils produced from pretreated pine wood. Appl Biochem Biotechnol 2009; 154(1–3):182–92.
[33] Shafizadeh F, Furneaux R, Cochran T, Scholl J, Sakai Y. Production of levoglucosan and glucose from pyrolysis of cellulosic materials. J Appl Polym Sci 2003;23(12):3525–39.
[34] Cheng S, Lin H, Chang S. Chemical composition and antifungal activity of essential oils from different tissues of Japanese cedar (Cryptomeria japonica). J Agric Food Chem 2005;53(3):614–9.
[35] Yatagai M, Makihara H, Oba K. Volatile components of Japanese cedar cultivars as repellents related to resistance to Cryptomeria bark borer. J Wood Sci 2002;48(1):51–5.
[36] Maiti S, Dey S, Purakayastha S, Ghosh B. Physical and thermochemical characterization of rice husk char as a potential biomass energy source. Bioresour Technol 2006;97(16):2065–70.